

SKPL-XXXX

SOFTWARE REQUIREMENTS SPECIFICATIONS

INTERLECT

For:

Miss Olaza

Prepared by:

Group “underrated”

Bachelor of Informatics Program

Faculty of Informatics

Telkom University

2025

 Program Studi S1 Informatika - Fakultas Informatika	SKPL - Nomor Dokumen		Halaman
	Revisi	<nomor revisi>	Tgl: <isi tanggal>

<i>Prodi S1 Informatika - Universitas Telkom</i>	<i>SKPL-xxx</i>	<i>Halaman 1 dari 26</i>
<i>Dokumen ini dan informasi yang ada di dalamnya adalah milik Prodi S1 Informatika-Universitas Telkom dan bersifat rahasia. Dilarang untuk mereproduksi dokumen ini tanpa diketahui oleh Program Studi S1 Informatika, Universitas Telkom</i>		

1) List of Changes

Revision	Description
A	Minimizing functional requirements from 24 to 11 functional requirement.
B	Recreate usecase diagram so it matches the requirement.
C	Reference using IEEE standard.
D	Merge student/user into student only.
E	Add elicition technique.
F	Humanize the one with AI.
G	Add a picture for UI reference.

INDEX	-	A	B	C	D	E	F	G
TGL		26 Oct. 2025	26 Oct. 2025	16 Nov. 2025	16 Nov. 2025	16 Nov. 2025	16 Nov. 2025	16 Nov. 2025
Ditulis oleh		Atallah	Atallah , Hafizh, Rakha	Rakha	Rakha	Rakha	Faizal	Atallah
Diperiksa oleh								
Disetujui oleh								

2) List of Changes Page

Page	Revision	Page	Revision
9	A		
10	B		
6	C		
7	D		
23	E		
20-24	F		
20-22	G		

3) Table of Contents

1) List of Changes	2
2) List of Changes Page.....	3
3) Table of Contents	4
1. Introduction.....	5
1.1 Purpose of Writing the Document	5
1.2 Scope of the Document.....	5
1.3 Definitions, Abbreviations, and Acronyms	5
1.4 References	6
2. Global Description of the Software	7
2.1 Statement of Objective of the Software.....	7
2.2 Perspective and Goals of the Software	7
2.3 User Profiles and Classes	7
2.4 Operating Enviroment	7
2.5 Software/System Limitations	8
2.6 Assumptions and Dependencies	8
3. Detailed Software Description	9
a) Needs Description.....	9
i) Functional Requirement	9
ii) Non-Functional Requirement.....	9
b) Analysis Modeling	10
3.2.1 Usecase Diagram.....	10
3.2.1.1 Usecase Scenario #1 “Login and Sign Up ”	11
3.2.1.2 Usecase Scenario #2 ”Search Universities”	12
3.2.1.3 Usecase Scenario #3 “Live consulting session with professionals”.....	14
3.2.1.4 Usecase Scenario #4 “AI Chatbot for consultation”	16
3.2.1.5 User Scenario #5 “Review and Rating”	17
3.2.2 Class Diagram:	19
4) External Interface Requirements	20
4.1 User Interface.....	20
4.2 Elicitation Technique	23
4.3 Software Interface	23
4.4 Communication Interface.....	24
4.5 intelligent System Features	24

1. Introduction

1.1 Purpose of Writing the Document

This document specifies the software requirements for **Interlect**, an international education finder software/application. This document will be used to guide the development and verification of the system, ensuring it meets the desired functional and non-functional requirements.

1.2 Scope of the Document

This Software Requirements Specification (SRS) covers the requirements for the **Interlect** application, which is designed to help students find their desired international education/college. The application's core functionality includes finding colleges based on a user's interest, current budget, and capability, as well as providing live consulting and AI-based education finding features. The system aims to inform students of their desired college and the necessary administration procedures.

1.3 Definitions, Abbreviations, and Acronyms

Term/Acronym	Definition
Interlect	The International Education Finder software/application.
SRS	Software Requirements Specification.
SKPL	Spesifikasi Kebutuhan Perangkat Lunak (Indonesian for Software Requirements Specification).
AI	Artificial Intelligence.
GPA	Grade Point Average.
IELTS	International English Language Testing System.
FR	Functional Requirement.
NFR	Non-Functional Requirement.
User	Refers to the Student user class.

Term/Acronym	Definition
Consultant	Refers to the Expert/Administrator user class.

1.4 References

[1] R. A. Wibowo, "Project Report Software Development and Analysis: International Education Finder Based on Interest, Budget, and Capability," **Unpublished Project Report**, School of Computing, Telkom University, Bandung, 2025.

[2] *Schoters*. (n.d.). Available: <https://www.schoters.com/id>. Accessed: Nov. 15, 2025.

[3] *ICAN-Education*. (n.d.). Available: <https://ican-education.com/>. Accessed: Nov. 15, 2025.

[4] *Telkom University*. (n.d.). Available: <https://telkomuniversity.ac.id/>. Accessed: Nov. 16, 2025.

[5] *Google Maps*. (n.d.). Available: <https://www.google.com/maps> . Accessed: Nov. 16, 2025.

[6] *Consolto*. (n.d.). Available: <https://www.consolto.com/features/live-chat> . Accessed: Nov. 16, 2025.

[7] *Bright Scholarship*. Available: <https://brightscholarship.com/> . Accessed: Nov. 16, 2025.

[8] *GSMarena*, Available: <https://www.gsmarena.com/> . Accessed: Nov. 16, 2025.

[9] *Dribbble*.. Available: <https://www.gsmarena.com/> . Accessed: Nov. 16, 2025.

2. Global Description of the Software

2.1 Statement of Objective of the Software

The main objective of **Interlect** is to help students find their desired college and understand the procedures required to pass the administration process. This product serves as an online international education finder to match students with colleges based on their specific inputs: interest (major, country, etc.), current budget (education and life expense, etc.), and capability (GPA score, IELTS score, etc.).

2.2 Perspective and Goals of the Software

Interlect is a **new, self-contained product** aimed at simplifying the complex process of searching for international education options. The core goals are to provide:

- A personalized university search based on multiple user criteria (interest, budget, capability).
- Access to live, professional consultation services (chat/video).
- AI-powered support for consultation and education finding.
- Comprehensive information including scholarship availability and estimated living costs.

2.3 User Profiles and Classes

The system identifies two primary user classes:

- **Student/User:**
 - **Profile:** Individuals seeking international education/college options.
 - **Role:** Primary users of the search, comparison, review, and consultation features.
 - **Requirements:** Must be able to log in/sign up, search for universities based on preferences, view details, give reviews, and engage in consultation sessions.
- **Consultant (Expert/Administrator):**
 - **Profile:** Professionals or experts in international education who provide guidance.
 - **Role:** Manages and conducts live consulting sessions via chat or video call.
 - **Requirements:** Must be able to log in/sign up and participate in live chat and video call sessions initiated by students.

2.4 Operating Environment

The software is an application intended to operate in an environment accessible by students and consultants. Given the nature of an international education finder, the system will likely operate on common platforms such as **web browsers** and/or **mobile operating systems (iOS/Android)**, requiring a stable **internet connection**. The system must interface with a **database** to store university, user, and consultant data. Furthermore, it integrates with a third-party service for **video conferencing (e.g., Zoom)** for live consultation sessions.

2.5 Software/System Limitations

Limitations that may affect the development or functionality include:

- **Data dependency:** The quality and quantity of university recommendations are limited by the information available in the system's Knowledge Base/Data.
- **AI capability:** The AI Chatbot may occasionally be unable to find a relevant or adequate answer for a user's query.
- **Consultant availability:** Live consulting sessions are dependent on a consultant being online and available.
- **Interface limitation:** The live video call feature is an alternative scenario during a live chat and relies on an external platform (e.g., Zoom).

2.6 Assumptions and Dependencies

Assumptions:

- Users (Students and Consultants) have access to a device (PC, tablet, or smartphone) capable of running the application and a stable internet connection.
- The system can accurately validate and verify user accounts during login/sign up.
- The AI Chatbot's Knowledge Base is maintained and continuously updated with relevant university, requirements, and scholarship data.

Dependencies:

- The Live Video Call feature is dependent on the successful integration with and functionality of an **external video conferencing service (e.g., Zoom)**.

3. Detailed Software Description

a) Needs Description

i) Functional Requirement

No.	Requirement Code	Description	Requirement Name
1.	FR-M1	System shall allow user to search university base on input (budget, interest, capability).	Search Universities
2.	FR-M2	System shall allow users to log in or sign up using their account credentials.	Login
3.	FR-S1	System shall display the scholarship availability.	Scholarship Availability
4.	FR-S2	System shall allow user to chat with AI Chatbot.	AI Chatbot
5.	FR-S3	System shall allow user to compare up several universities side-by-side.	Compare Universities
6.	FR-S4	System shall display estimated living cost for each country selected.	Estimate Living Cost
7.	FR-S5	System shall allow user to have live consulting session with consultant via chatting.	Live Consulting
8.	FR-S6	System shall allow user to view partnership available universities with the software.	Partnership availability
9.	FR-S7	System shall allow user to request for a video call with consultant.	Video Call with Consultant
10.	FR-D1	System shall display university recommendation base on previous search by user.	University Recommendation
11.	FR-D2	System shall provide review and rating section for each university from past students.	Review and Rating

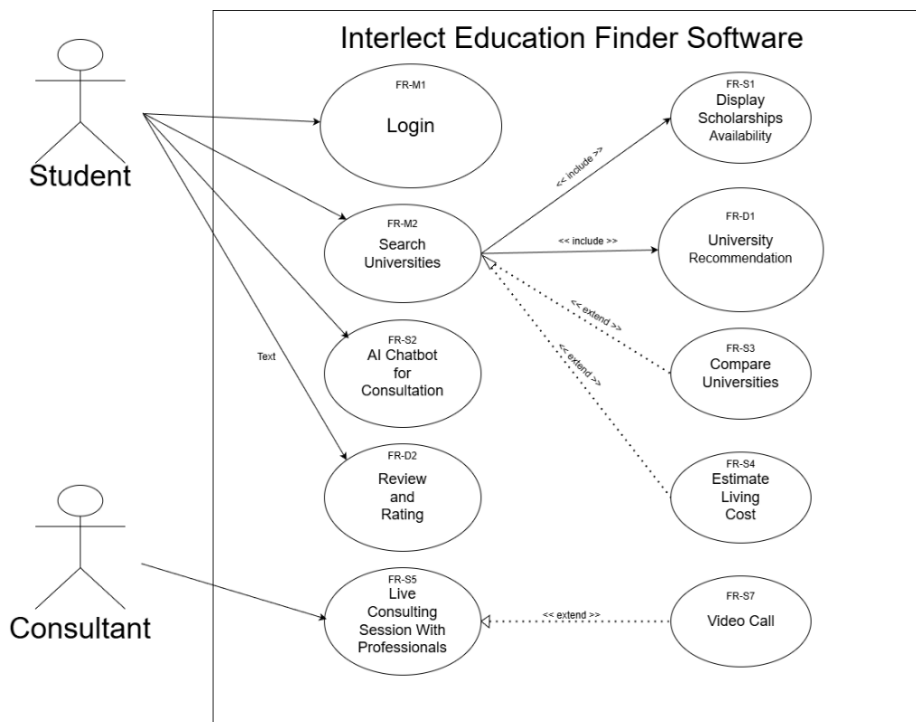
ii) Non-Functional Requirement

No.	Quality Criteria	Requirement Code	Description
-----	------------------	------------------	-------------

1.	Performance	NFR-PE1	The system must display university search results based on user preferences (budget, interest, capability) within a maximum of 3 seconds .
2.	Performance	NFR-PE2	The system must support at least 1,000 concurrent users without significant degradation in processing speed, especially during peak consulting or search hours.
3.	Security	NFR-SE1	All user account credentials (email, password) must be secured using strong encryption and hashing techniques.

b) Analysis Modeling

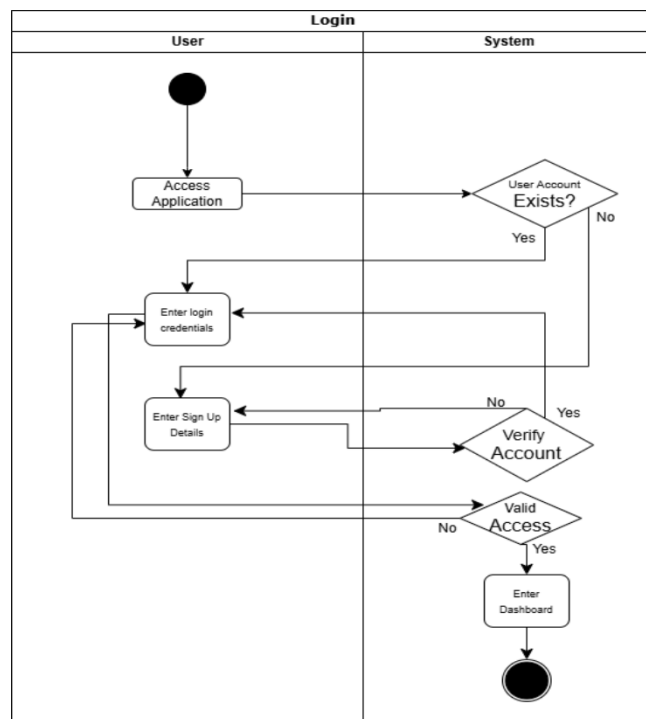
3.2.1 Usecase Diagram



3.2.1.1

Usecase Scenario #1 “Login and Sign Up ”

Use Case Name	Logging In	
Description	User/Admin can log in with owned account	
Pre-Condition	User/Admin have an account	
Post-Condition	System will give access to user/admin and move to home screen	
Main Scenario		
	Actor	System
	1. Logging in	
		2. Displaying Login Form
	3. Inputting Email and Password.	
	4. Click the login button	
		5. Validating Email and Password
		6. Transition into main/home screen
Alternative Flow		
	Actor	System
	1. Logging in.	
		2. Displaying Login Form
	3. Clicking sign up option.	
		4. Displaying Sign up form
	5. Signing up using manually, or by using other accounts	
	6. Inputting Email, Username, and Password.	
	7. Clicking sign up button	
		8. Saving username, email, and password to the system database

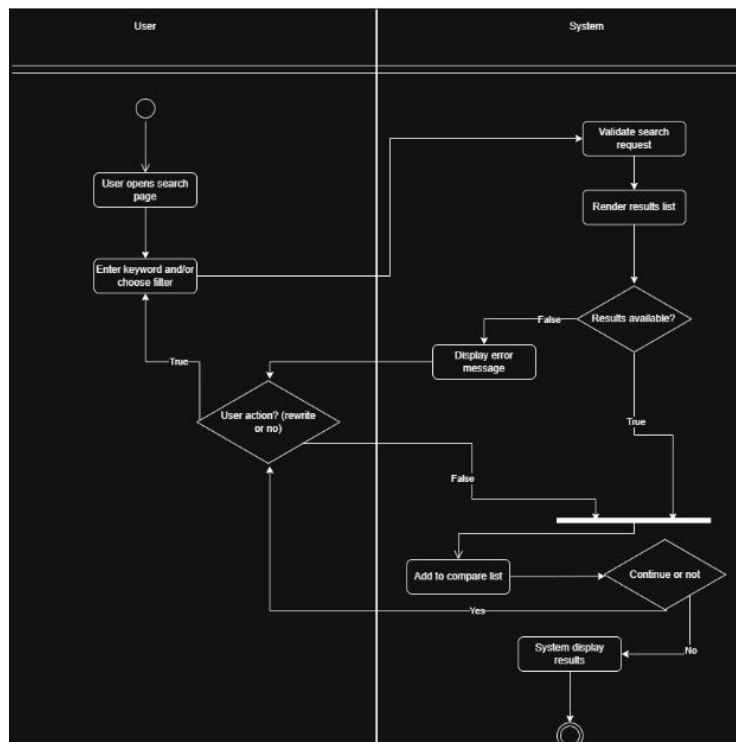


3.2.1.2

Usecase Scenario #2 "Search Universities"

Use Case Name	Search Universities	
Description	The user can generate the international school or college based on user inputs (interest, budget, capability)	
Pre-Condition	User have fulfilled the data needed	
Post-Condition	The system will generate matching university based on user preferences	
Main Scenario		
	Actor	System
	1. User clicks the generate university	
		2. Displayed new page filled with data that needs to be inputted
	3. User input data based on interest, budget, and capability	
	4. User Clicks Generate	

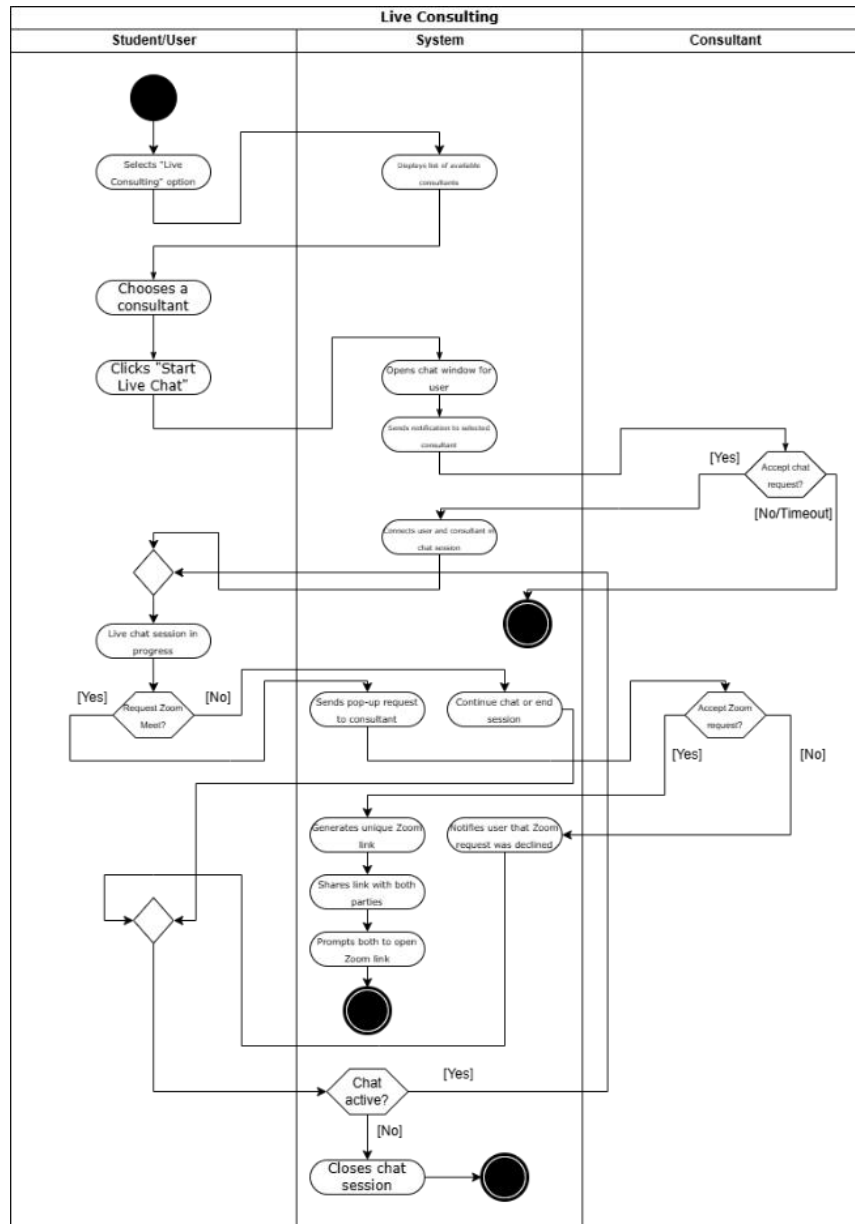
		5. System will generate universities based on user input
	6. User can compare or view university details for further information	
		7. System will provide information about the provided university, whether it is to be compared or read
	Alternative Flow	
		8. The System cannot find the desired university and will display apologizing message



3.2.1.3

Usecase Scenario #3 “Live consulting session with professionals”

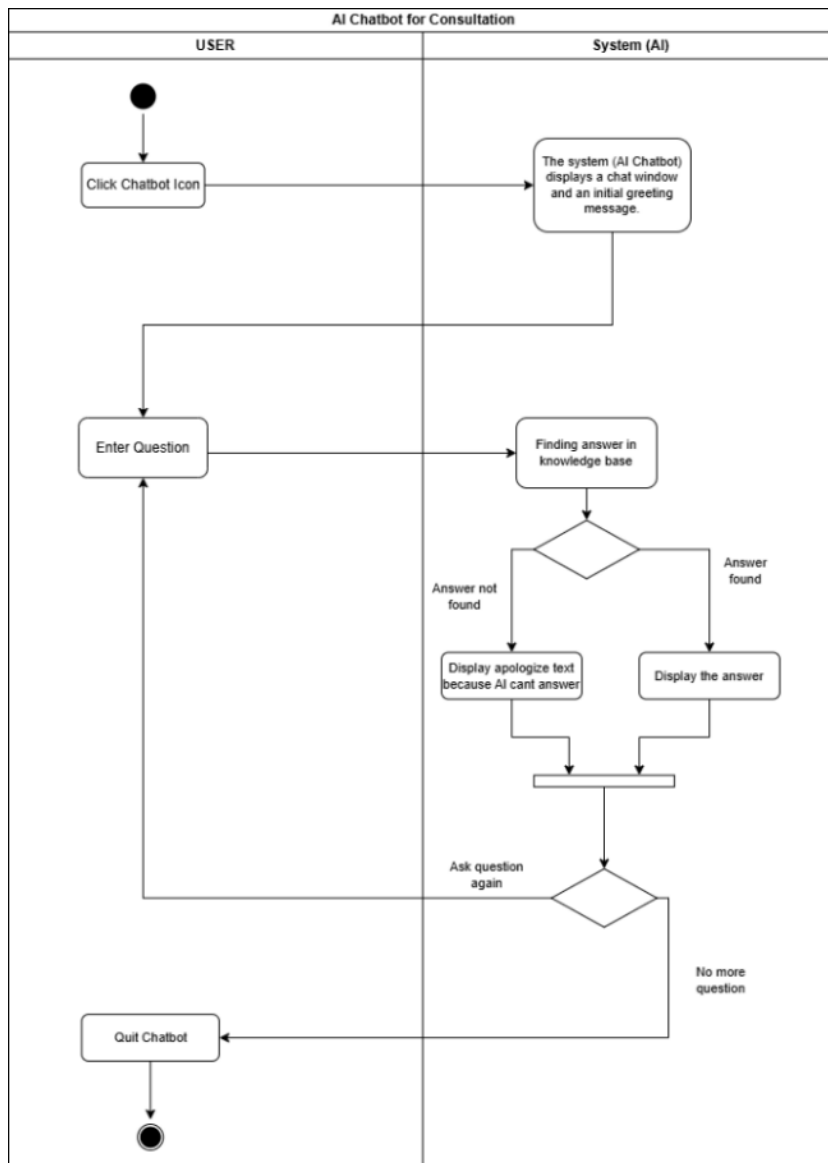
Use Case Name	Live Consulting session with professionals	
Description	The user can have a live consulting session with an international education consultant regarding the user preferences	
Pre-Condition	User and admin have questions ready for the consultation session	
Post-Condition	The international consultant will give college recommendation based on user answers	
Main Scenario		
	Actor	System
	1. User clicks the Live Consultation	
		2. Display list of available consultant
	3. User can freely choose consultant	
		4. Opens chat window for user
	5. Use can also have a live call session with the consultant, or chat session according to the consultant agreement	
		6. Connects user and consultant in chat session



3.2.1.4

Usecase Scenario #4 “AI Chatbot for consultation”

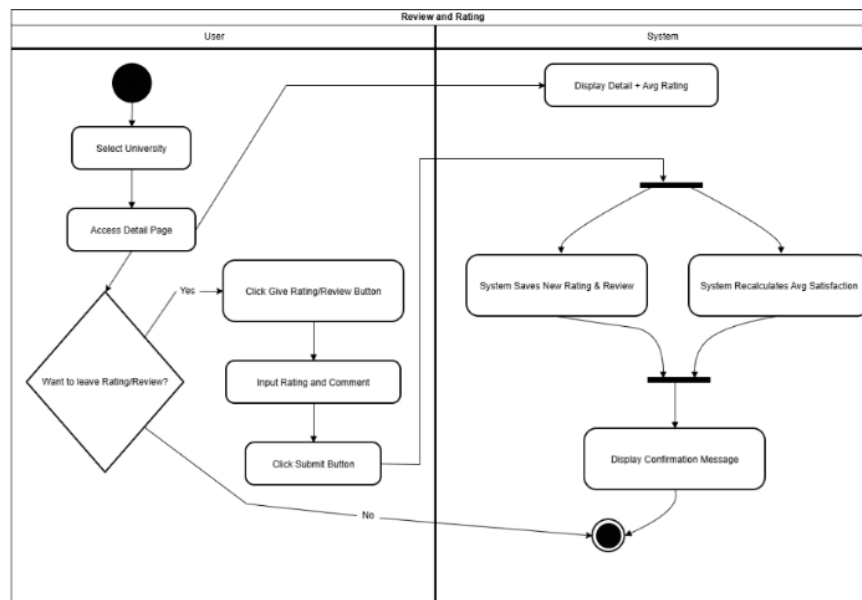
Use Case Name	AI Chatbot for Consultation	
Description	The user can have a live consulting session with an AI that have a purpose of giving insights and answers to all the users questions	
Pre-Condition	User have fully understood the ethics of AI usage, especially in the education sector	
Post-Condition	The education AI will give answers based on student’s questions, with data and descriptive explanation	
Main Scenario		
	Actor	System
	1. User clicks the AI Chatbot option	
		2. The AI will greet the user and ask if there’s anything they want to ask,
	3. The user can give questions, via texting or voice note	
		4. The AI will give answers based on the user questions, with available data and descriptive explanation
	5. User can still give extra regarding education	
		6. The AI will keep giving answers until the user is finished
	Alternative Flow	
		7. The AI cannot find the university and thus the system will display apologizing message



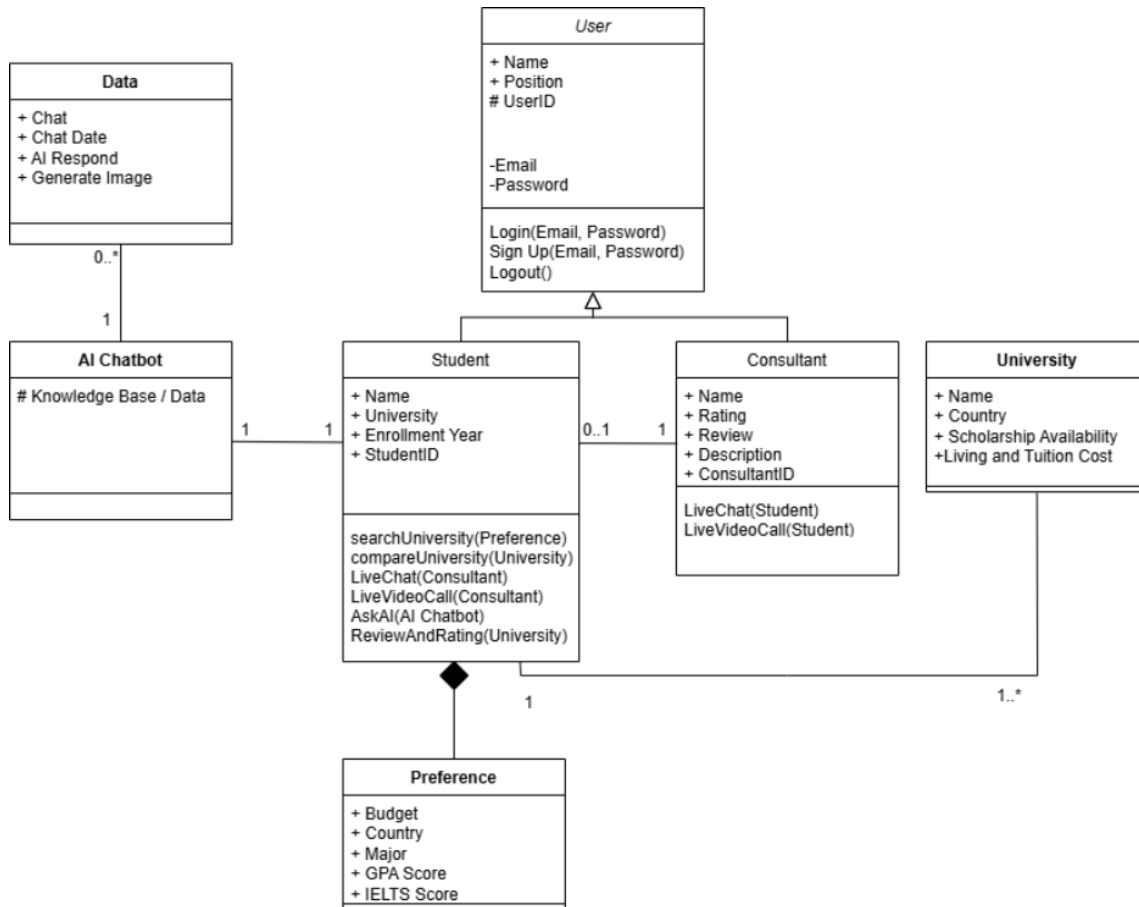
3.2.1.5 User Scenario #5 “Review and Rating”

Use Case Name	Review and Rating	
Description	User can give or check the review and rating score on every university.	
Pre-Condition	User have fully aware of the desired university quality, and user can filter the general review or the personal review	
Post-Condition	User can know what to expect from the desired university, based on others review and ratings	
Main Scenario		
	Actor	System
	1. User clicks on the university details	

		2. The system will display the universities along with the review and ratings
	3. The user can check the review and rating from every other user	
	4. The user can also give review and ratings	
		5. The system will display the review and rating form to the user, and later will store the information to be shared across the platform
	Alternative Flow	
	6. User decide not to give review and rating	



3.2.2 Class Diagram:



4) External Interface Requirements

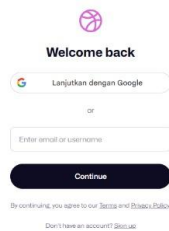
4.1 User Interface

Student (User) and Consultant (Expert / Administrator) are the two main user types in the system. The unique workflows of each classes must be taken into account when designing the User Interface (UI).

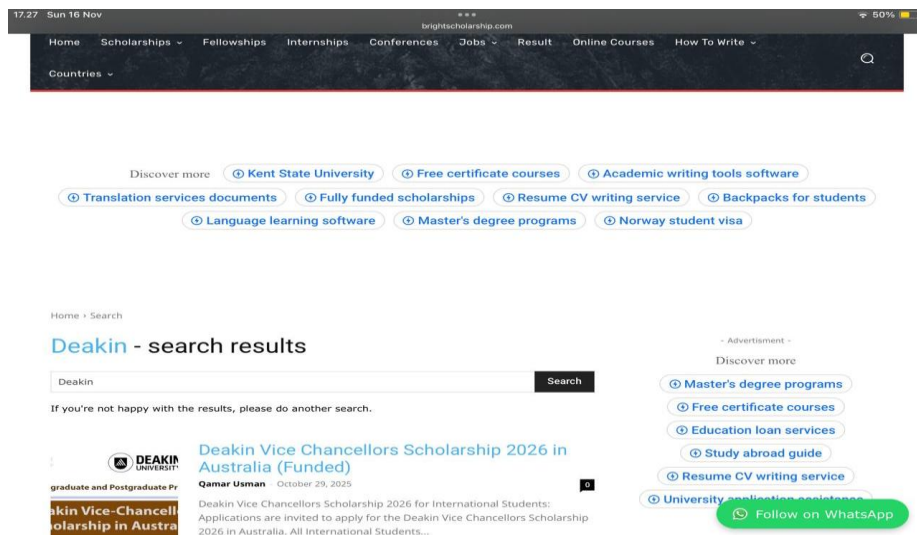
Student (User) Interface:

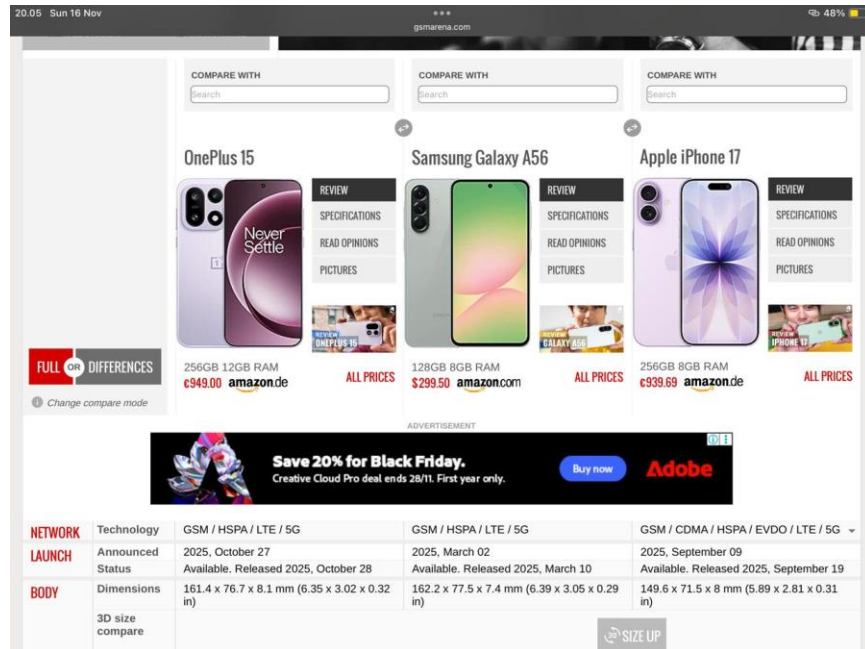
- The UI must provide a clear flow for Login and Sign Up, including forms for "Inputting Email and Password"[9].

Dribbble

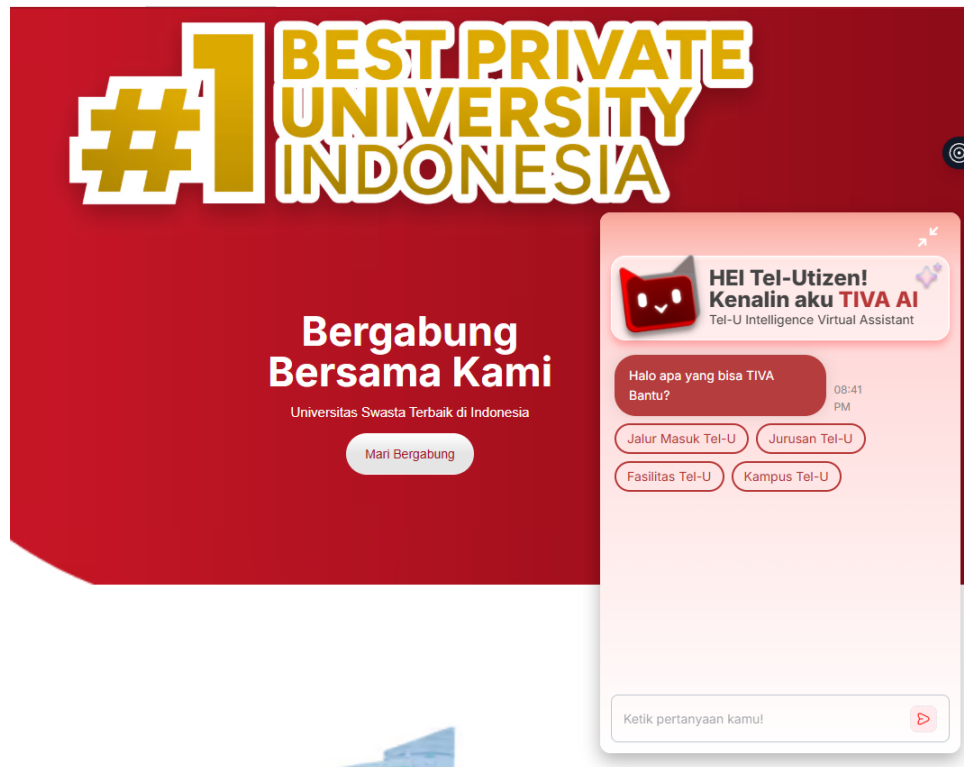


- The UI must feature a University Search page allowing the user to "Enter keyword and/or choose filter" based on interest, budget, and capability [7][8].

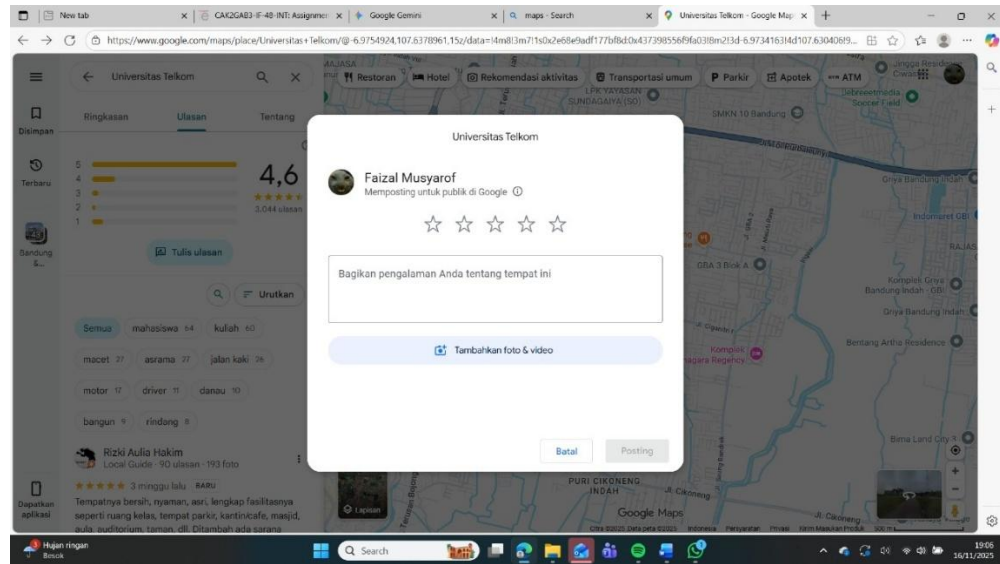




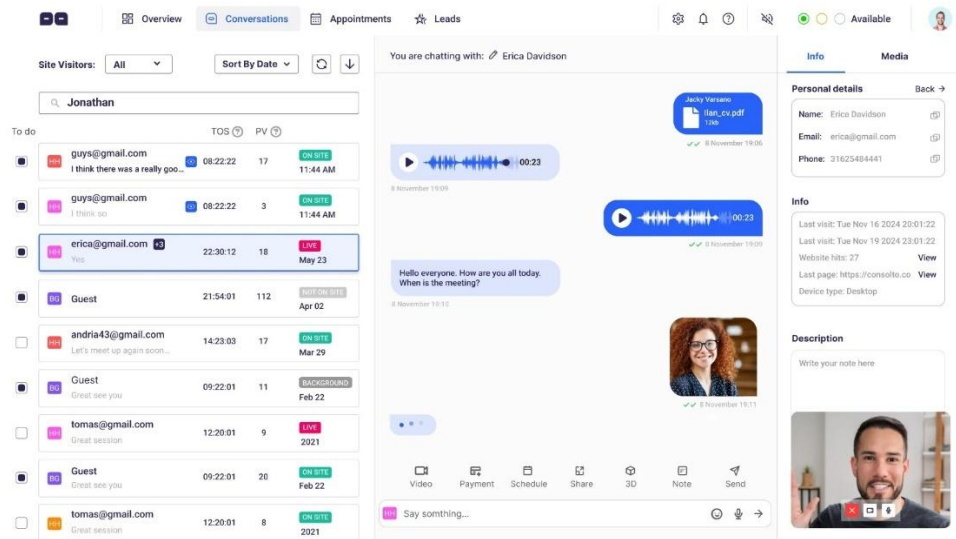
- The UI must provide an intuitive chat interface for the AI Chatbot (FR-S2), allowing the user to "Enter Question"[4].



- The UI must clearly display university details and provide functionality for the user to view and "Click Give Rating/Review Button" on the Review and Rating page[5].



- The UI must provide a clear button or link for the user to "Selects 'Live Consulting' option" and "Chooses a consultant" to initiate a Live Consulting session (FR-S5)[6].



- **Consultant (Expert/Administrator) Interface:**

- The UI must provide a dashboard for the consultant to receive and manage chat sessions.
- The UI must allow the consultant to "Accept chat request" and respond to a "Request Zoom Meet?" from the user.

4.2 Elicitation Technique

Requirement Type	Elicitation Type	Technique Used	Target	Summary
Must Have	Observation	Field Observation	International consultant	The consultant will giving and asking a question based on student who are consulting
Satisfier	Survey	Questionnaire	students	Students are asked to fill out the questionare to
				obtain brief information from the student, and they are able to provide feedback for the future
Delighter	Creative	Brainstorming	Group member	Each Group ember involves generating ideas allowing for free flowing thought

4.3 Software Interface

- **Database System (Internal):**
 - The system will interface with an internal database for "Validating Email and Password" during the Login process.

- The database will be used for "Saving password to the system database" during Sign Up.
- The database will be used to store and retrieve university data for Search Universities and to "System Saves New Rating & Review".
- **AI Knowledge Base API:**
 - For the AI Chatbot (FR-S2), the system will interface with a software API for its knowledge base to enable "Finding answer in knowledge base".
- **Third-Party Video Conferencing API (e.g., Zoom):**
 - For the Video Call (FR-S7) feature, the system will interface with an external service's API (such as Zoom, as indicated in the activity diagram).
 - This interface must be able to "Generates unique Zoom link" and "Shares link with both parties".

4.4 Communication Interface

- **HTTPS (HTTP Secure):**
 - All standard communication between the client application (web/mobile) and the backend server must use the HTTPS protocol to ensure data security, especially for sensitive processes like Login (FR-M2), Sign Up, and user data transmission.
- **WebSocket:**
 - For real-time communication features like Live Consulting (FR-S5) and AI Chatbot (FR-S2), the system will use the WebSocket protocol.
 - This is required to facilitate the persistent, two-way "chat session" between the user and consultant, and between the user and the AI.
- **API Calls (HTTPS):**
 - Communication with third-party services, such as the Zoom API for creating video calls, will be conducted via secure HTTPS API calls.

4.5 intelligent System Features

1. **AI Chatbot for Consultation (FR-S2):**
 - a. The system will provide "AI based education findings".
 - b. This feature will use Natural Language Processing (NLP) to understand user questions and provide answers from a knowledge base. The system must also intelligently handle cases when an "Answer not found".
2. **University Recommendation (FR-D1):**
 - a. The system will provide a smart feature to "display university recommendation base on previous search by user".
 - b. This requires an algorithm to analyze the user's search history and preferences to suggest relevant universities.

Appendix A: List of Foreign Words

Term/Acronym	Definition
Interlect	The International Education Finder software/application.
SRS	Software Requirements Specification.
SKPL	Spesifikasi Kebutuhan Perangkat Lunak (Indonesian for Software Requirements Specification).
AI	Artificial Intelligence.
GPA	Grade Point Average.
IELTS	International English Language Testing System.
FR	Functional Requirement.
NFR	Non-Functional Requirement.
User	Refers to the Student user class.
Consultant	Refers to the Expert/Administrator user class.

Member	Contribution
Rakha	Global Description of the Software, class diagram, usecase scenario & activity diagram for Login.
Atallah	Fungtional Requirement, usecase diagram, usecase scenario & activity diagram for “AI Chatbot”.
Faizal	External Interface Requirement, usecase scenario & activity diagram for “Review and Rating”.
Hafizh	Usecase Diagram, usecase scenario & activity diagram for “Live Consultation”.
Gavin	Introduction, usecase scenario & activity diagram for “Search Universities”.